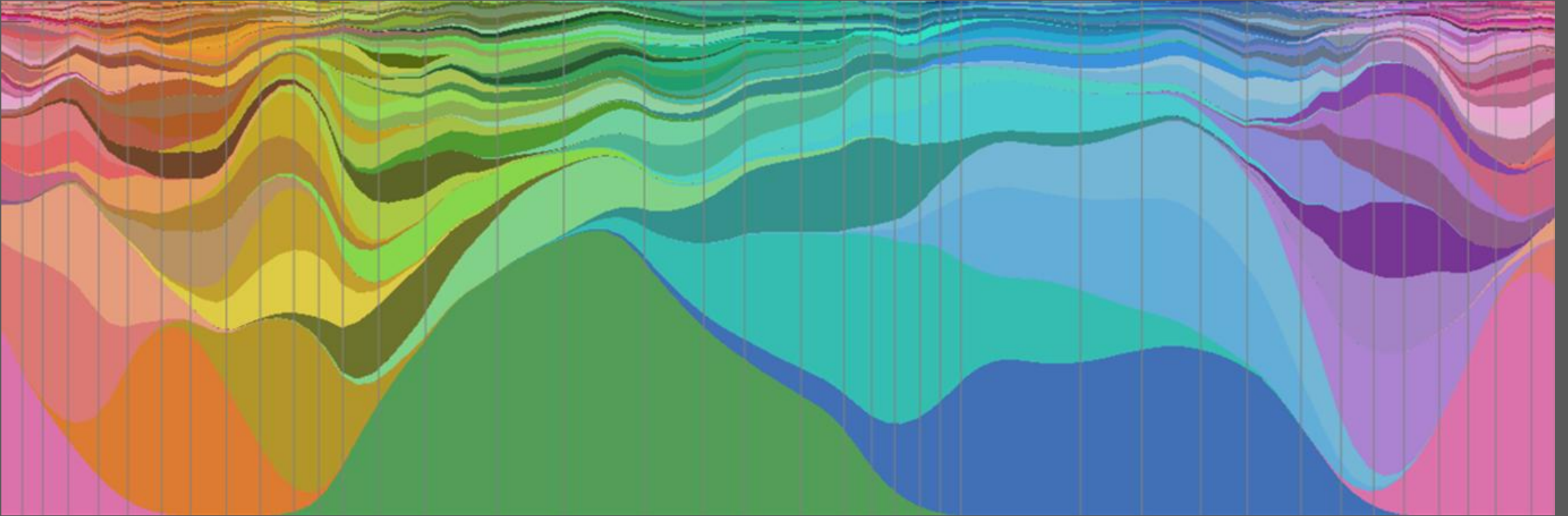


LDSCI5209 - Information Presentation and Visualisation

Week 6: Maps and Scientific Visualisations



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Northeastern University London

Recap: interaction and animation

- Personas
- Interaction
- Animation

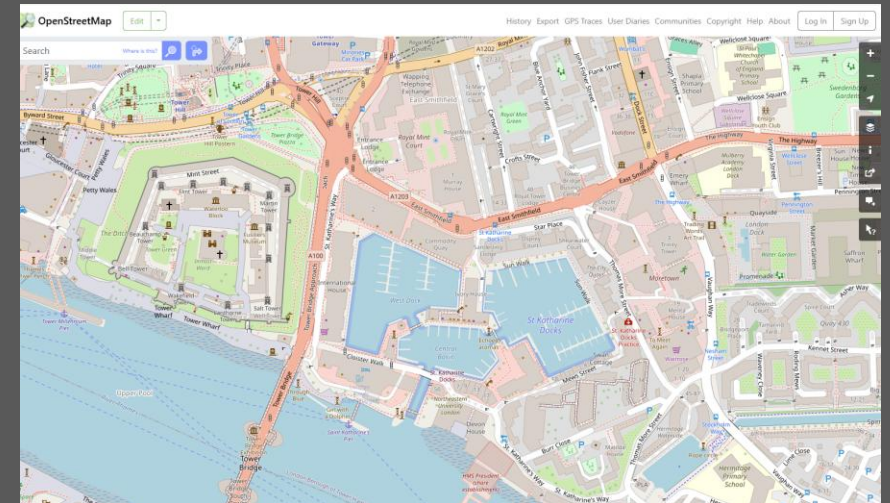
Plan for today

- Map types
- Projections
- Scientific visualisations

Maps

Types, projections, mapping methods, resources

Cartography



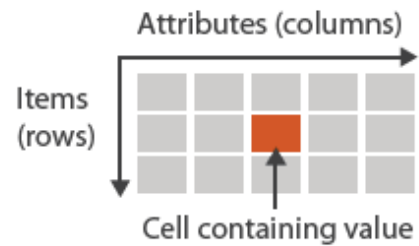
London, Open Maps (2025)

Ptolemy's Geography (circa 150)

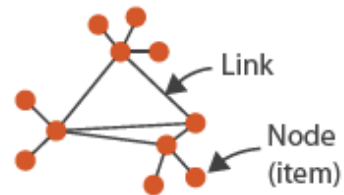
Spatial data types

➔ Dataset Types

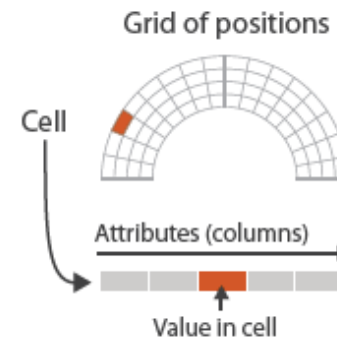
➔ Tables



➔ Networks



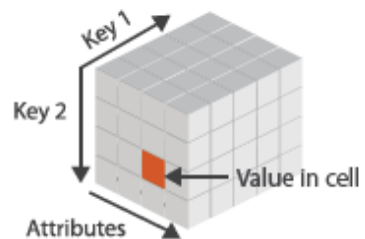
➔ Fields (Continuous)



➔ Geometry (Spatial)



➔ Multidimensional Table



➔ Trees



Arrange Spatial Data

➔ Use Given

➔ Geometry

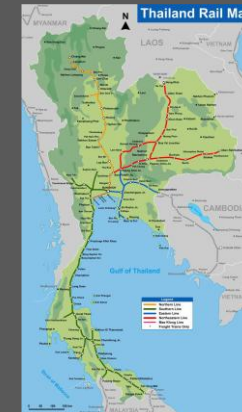
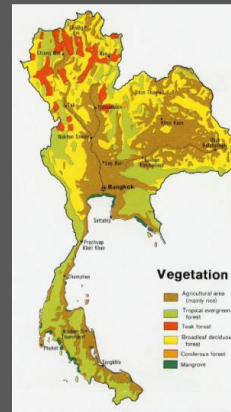
➔ Geographic

➔ Other Derived

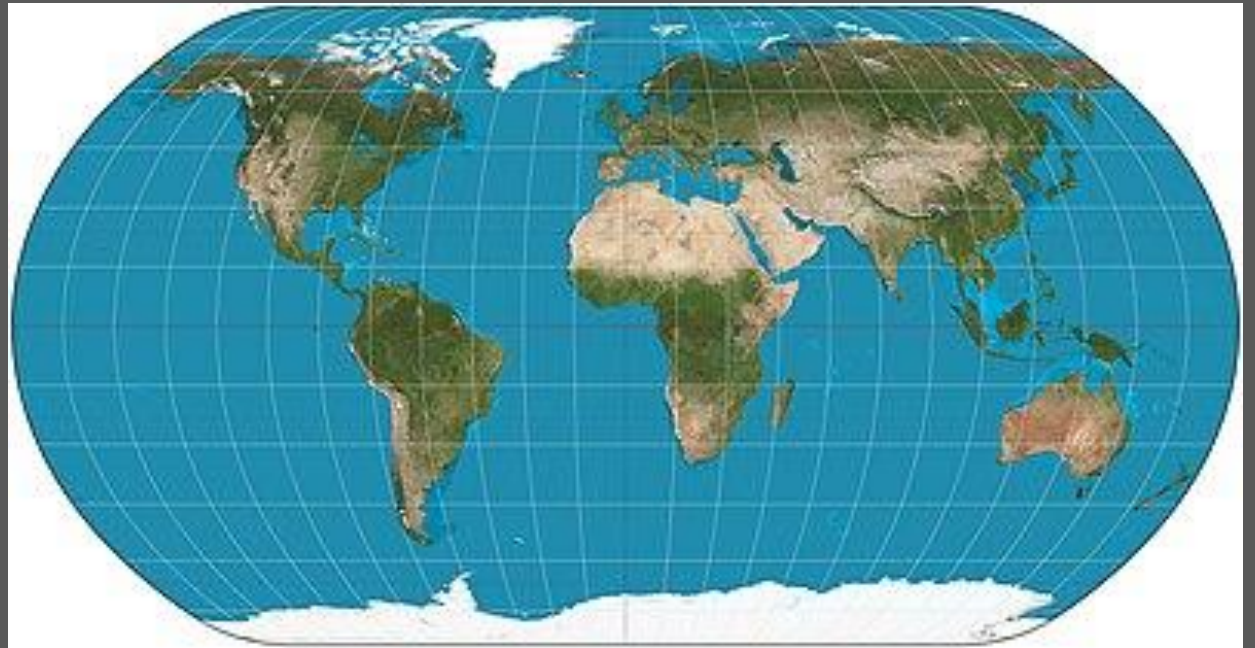
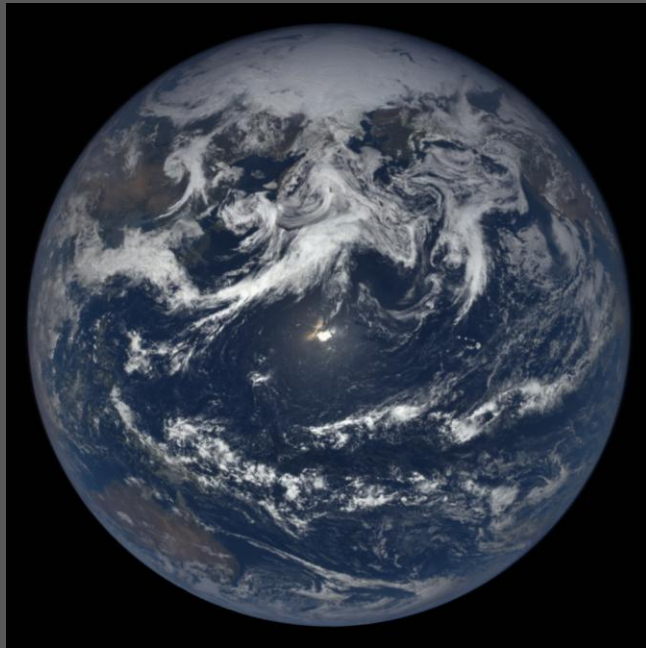


Types of Geographic Maps

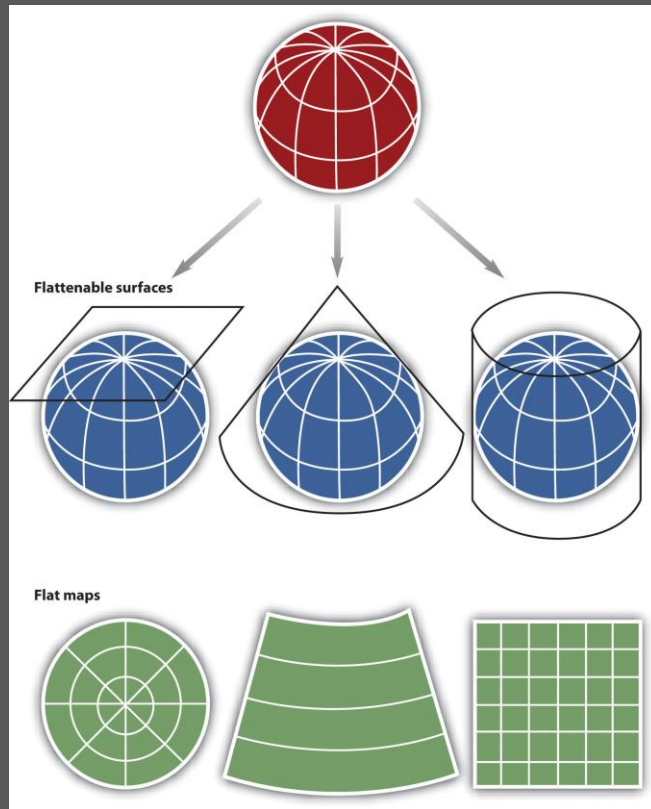
- **Reference maps:** General geographic context without emphasis on specific data.
- **Thematic maps:** Highlight spatial variation of a specific theme or variable.
- **Special-purpose maps:** Task-specific thematic maps for specialised use.



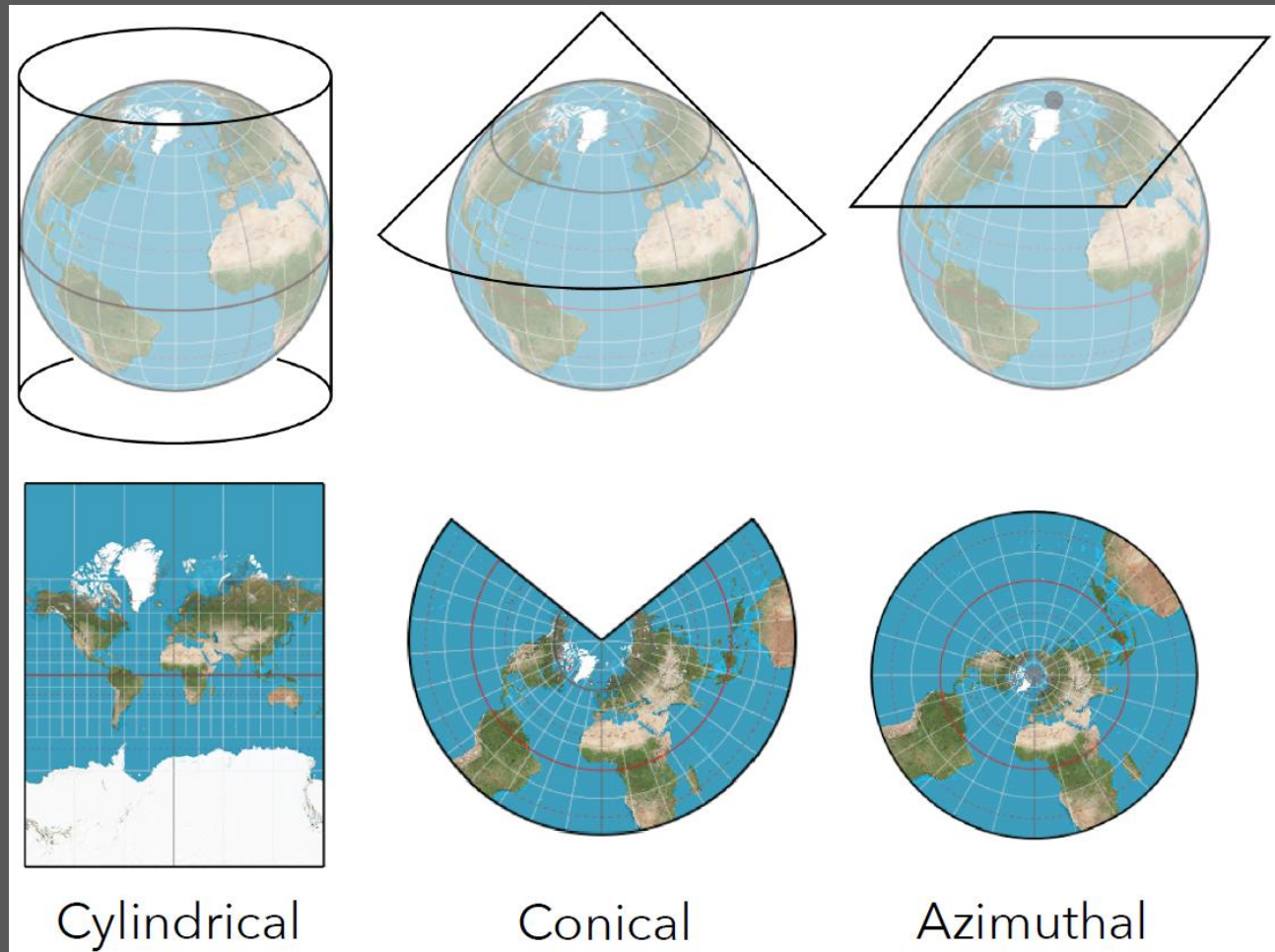
Map projections



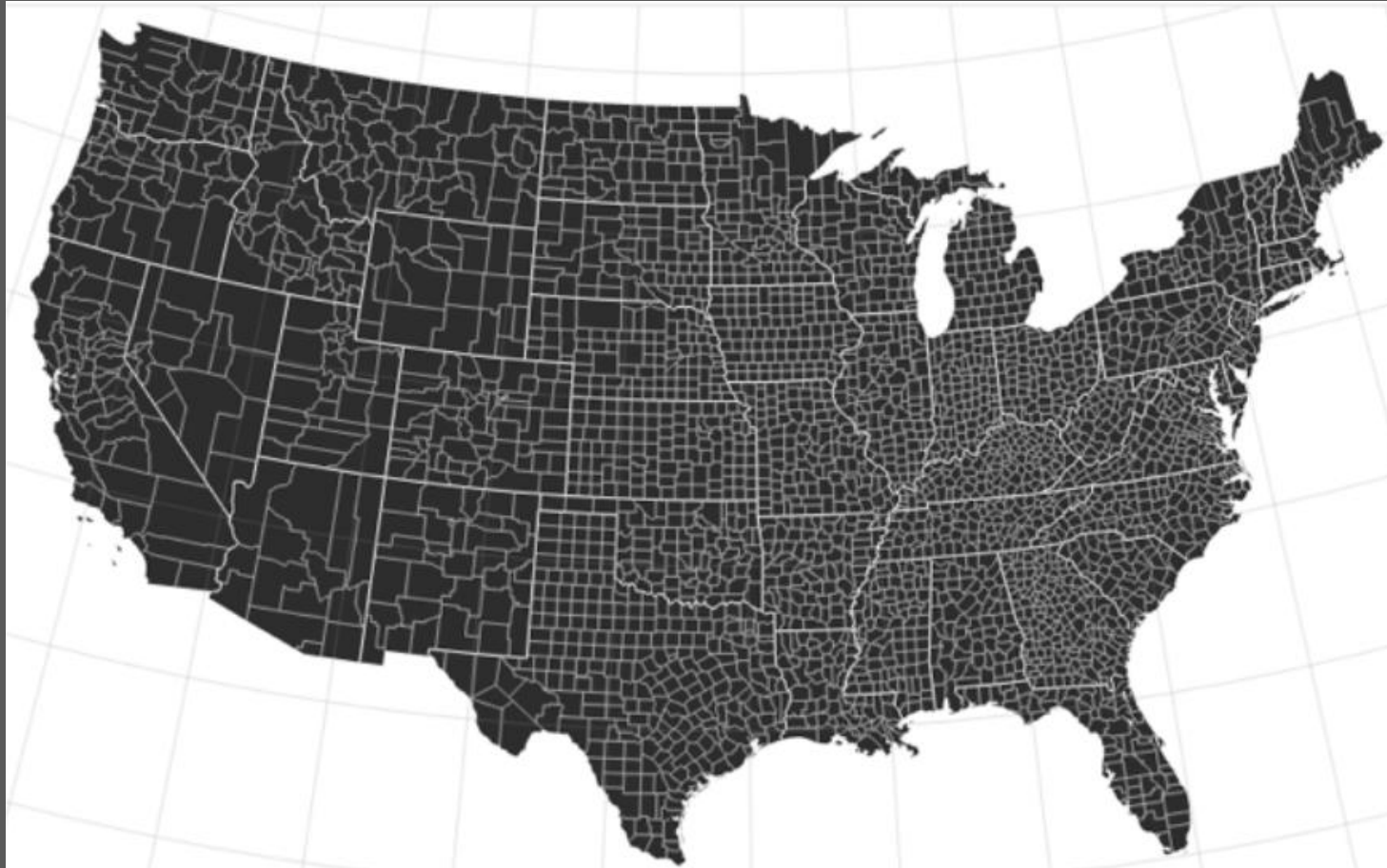
Projection types



https://en.wikipedia.org/wiki/List_of_map_projections



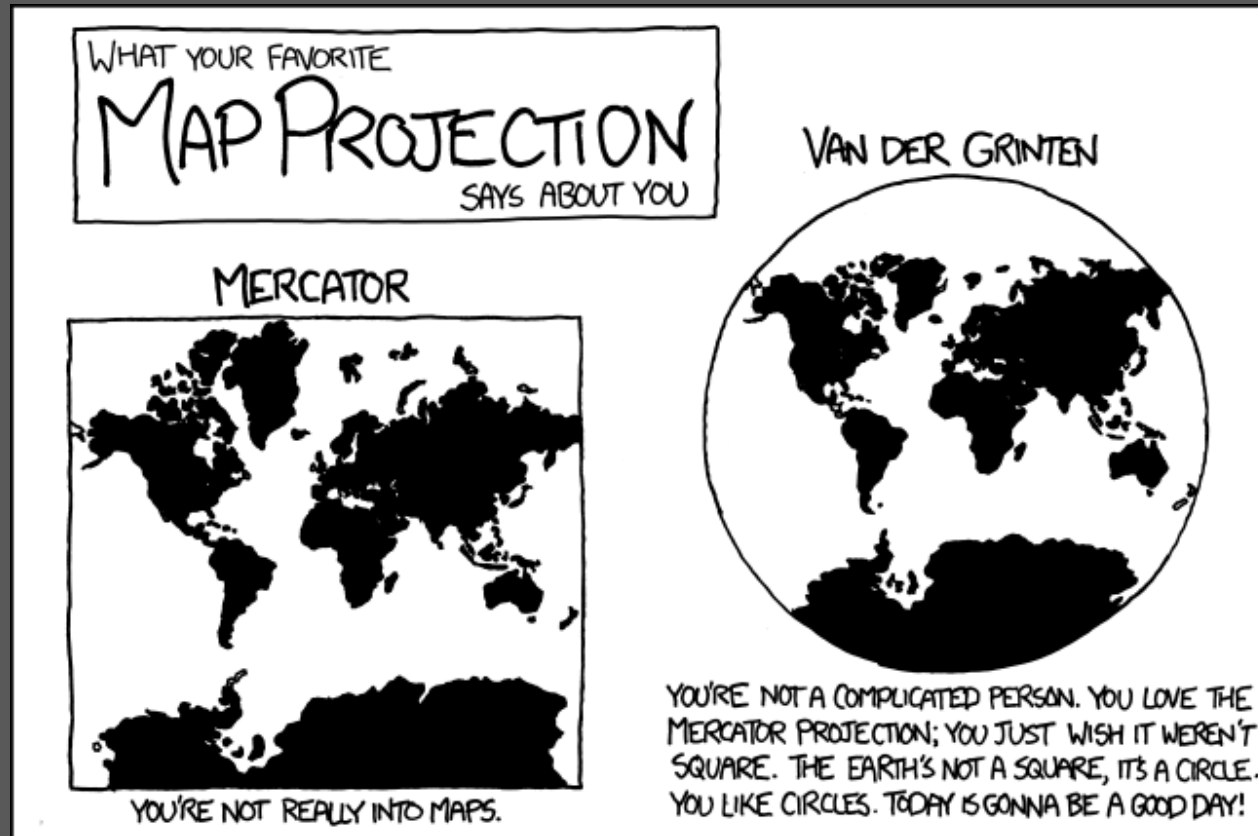
Albers equal-area conic



Spherical Mercator

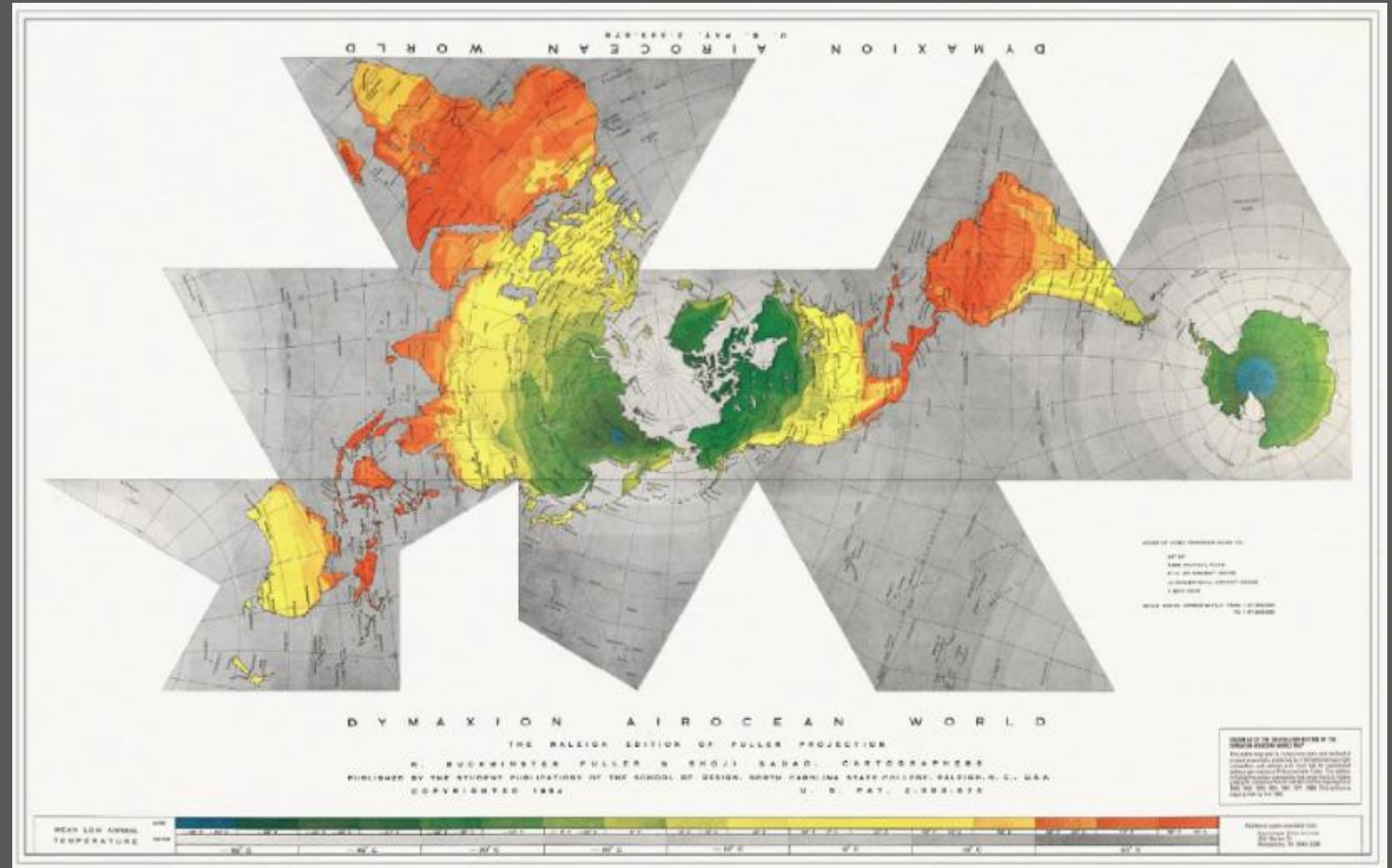
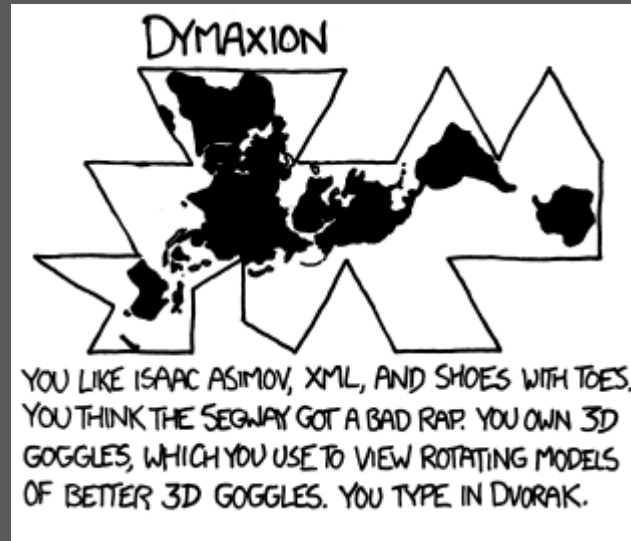


Square or circle projection?



<https://xkcd.com/977/>

Dymaxion projection



Mapping data

- Symbol Maps → plot data over a map
- Choropleth Maps → coloured regions
- Heatmaps & Contours → show densities
- Cartograms → distort to show quantities
- Flow Maps → flux across regions

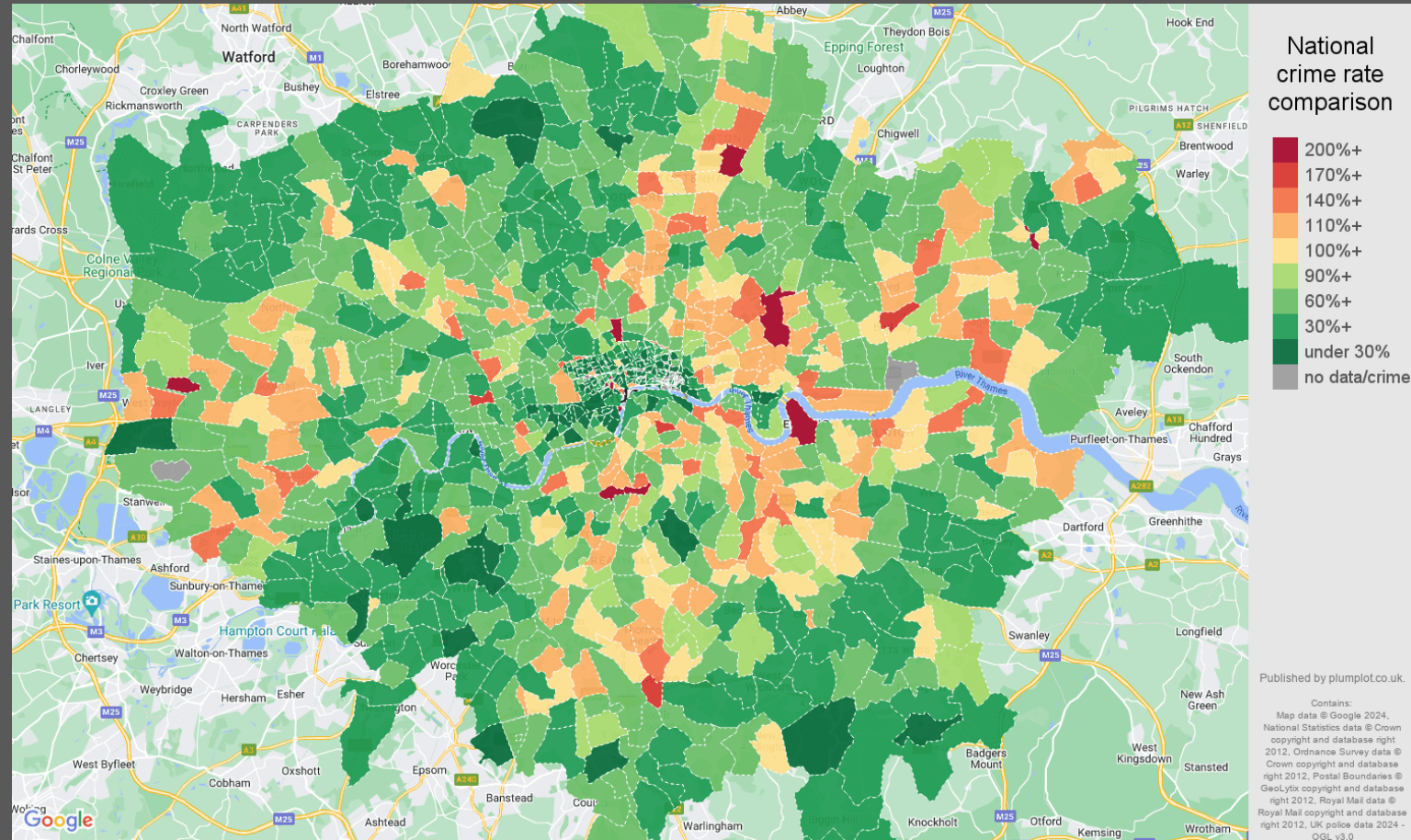
London symbol map



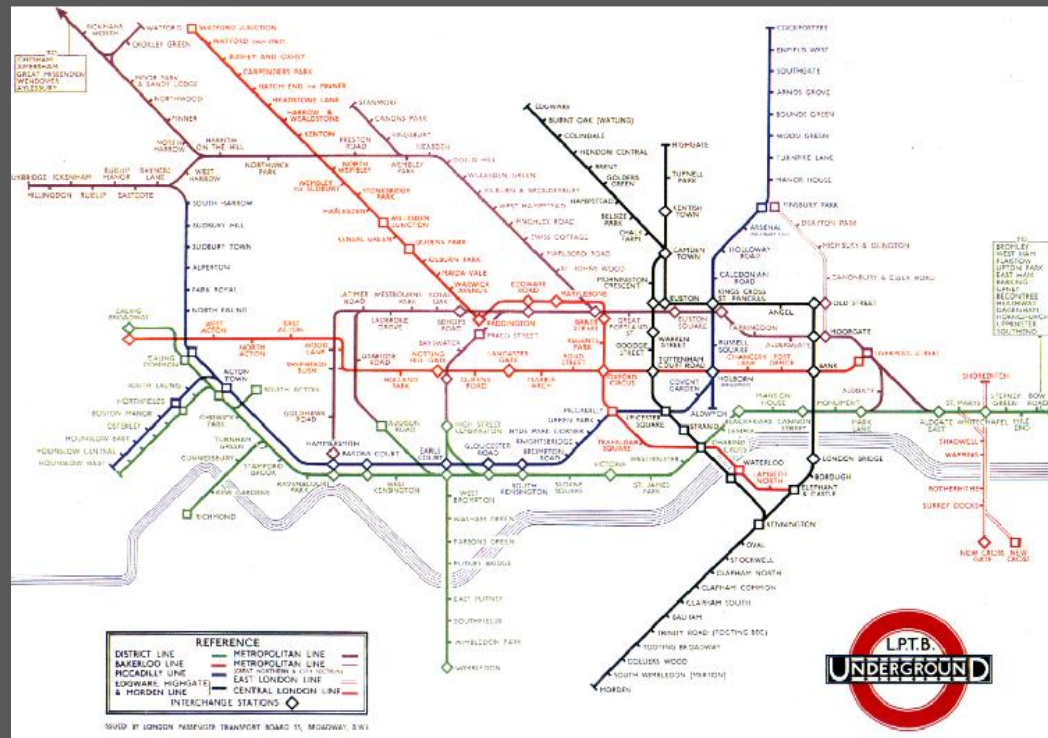
London choropleth map



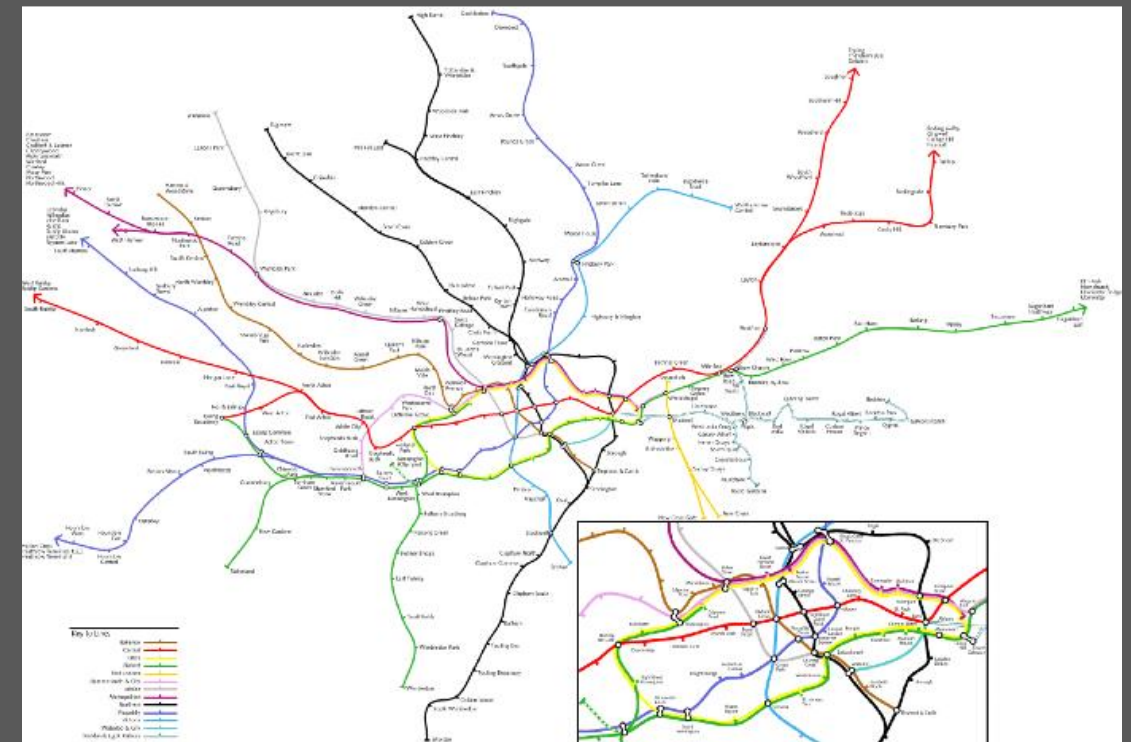
London crime rate contour map



London tube (route) map

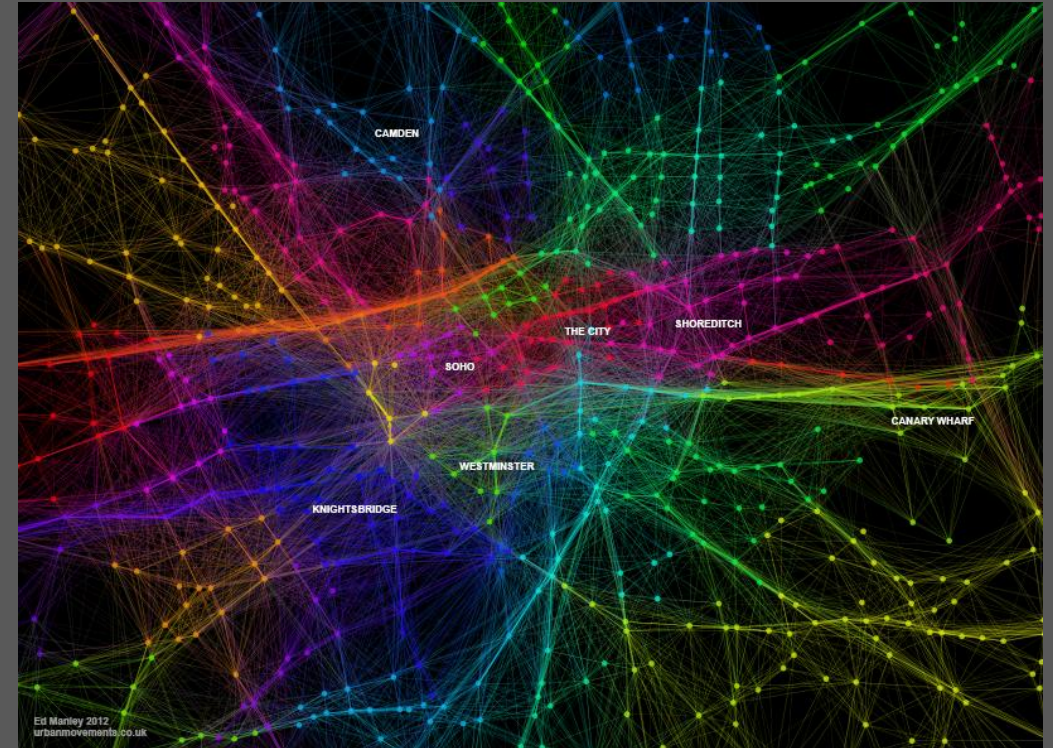
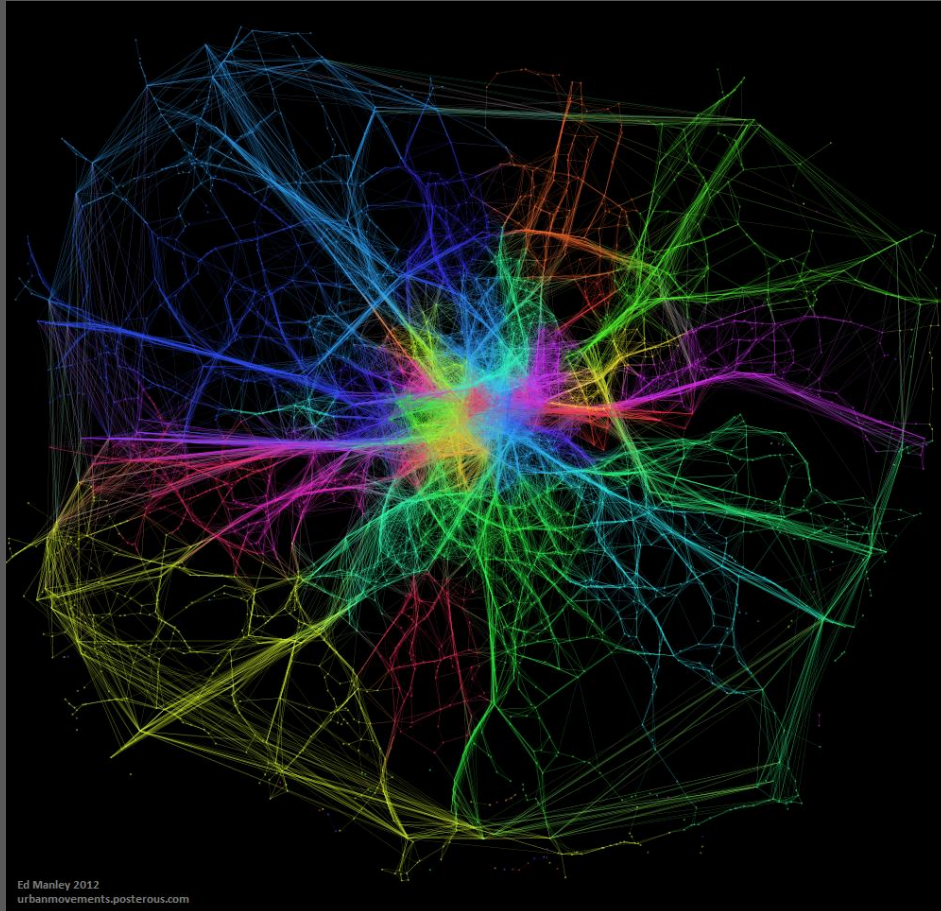


London Underground (Beck 33)



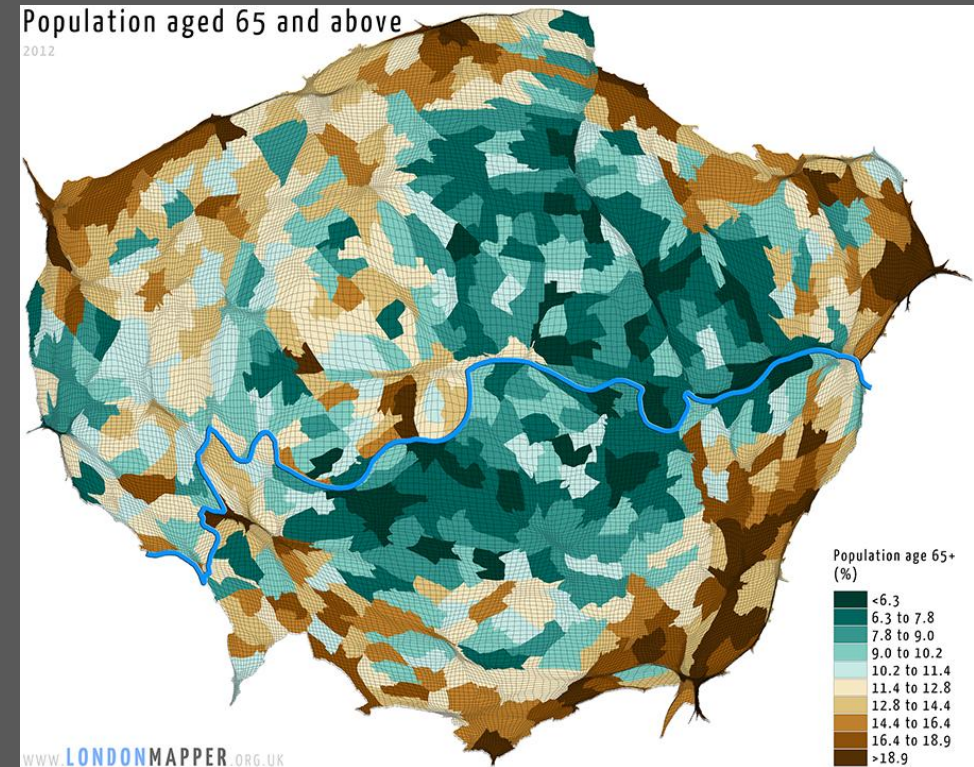
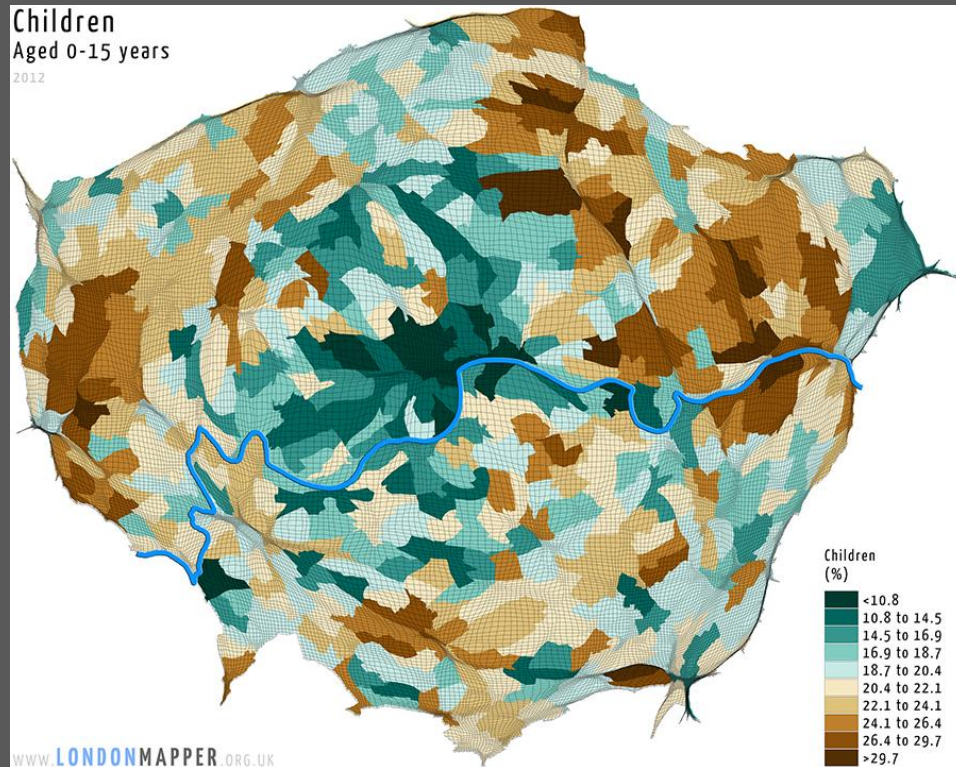
Geographic version of map

London traffic flow



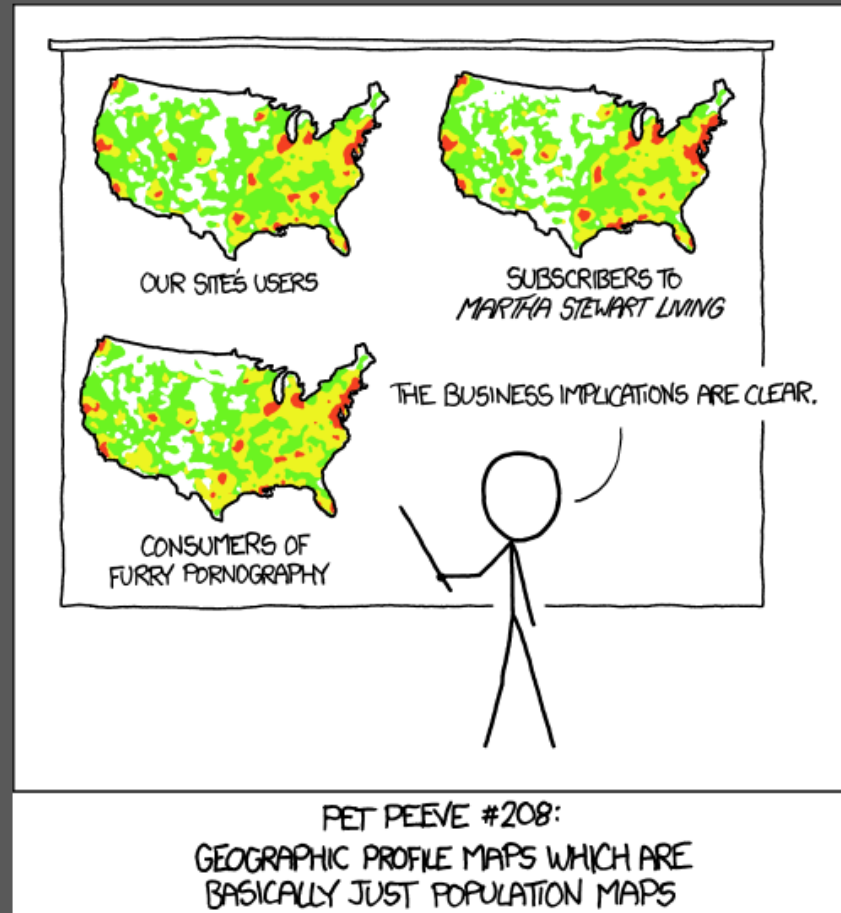
<http://urbanmovements.co.uk/tag/london/>

London cartograms



<https://www.theguardian.com/uk-news/datablog/gallery/2014/may/11/social-dimensions-shape-london-mapped>

Maps can also mislead...



<https://xkcd.com/1138/>

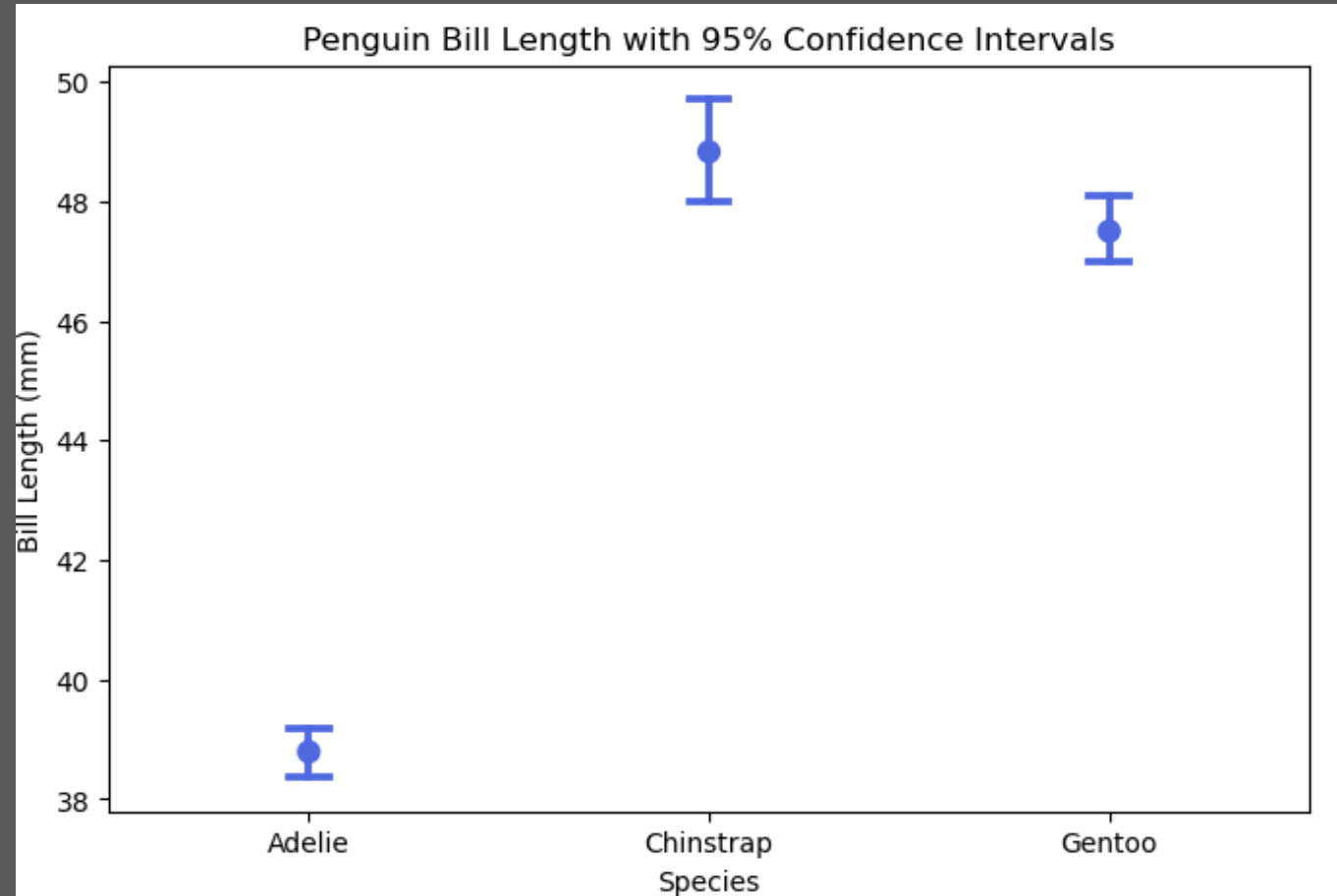
Scientific visualisations

Uncertainty, confidence intervals, error bars, density plots, bootstrapping

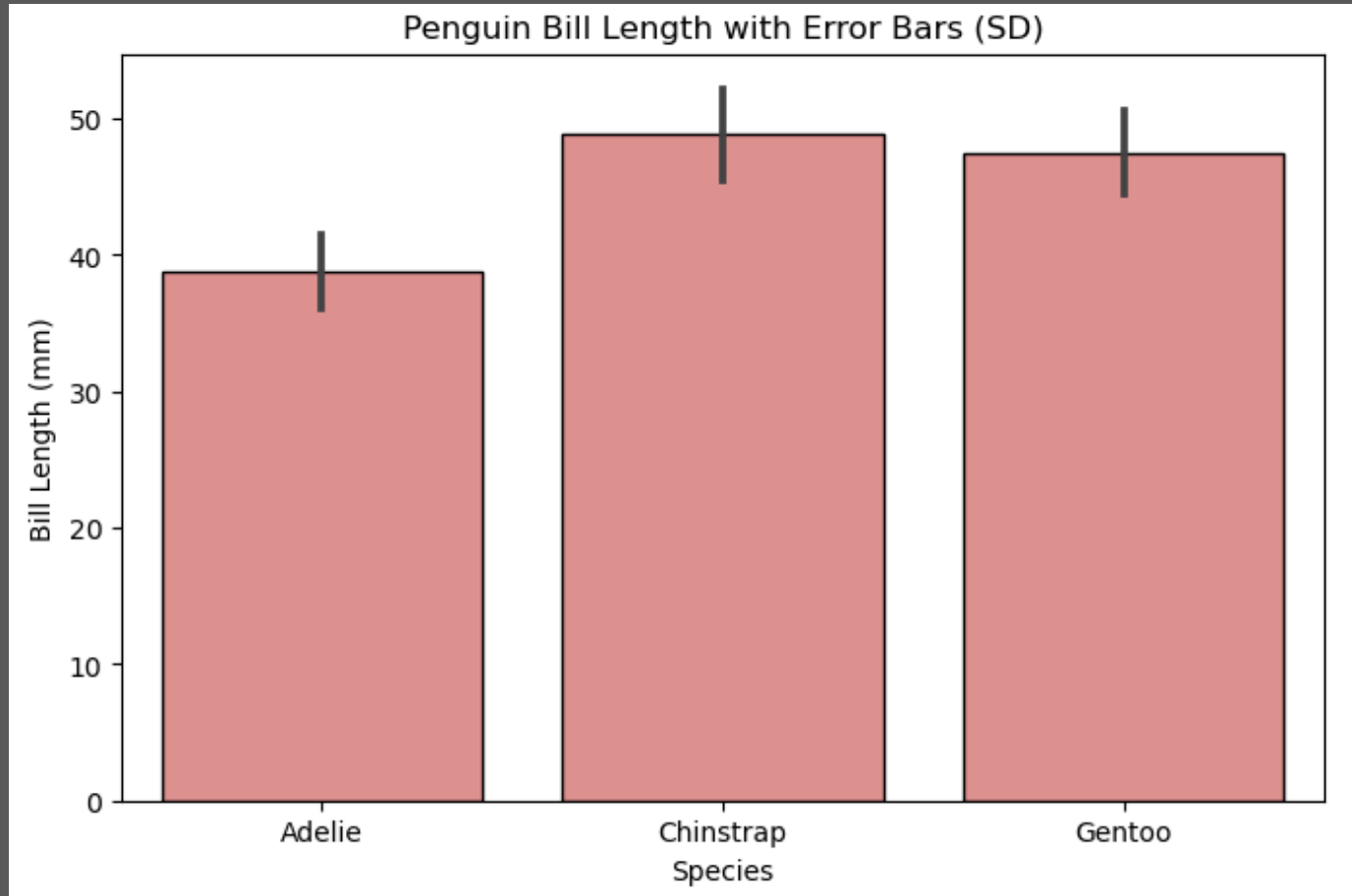
Visualising uncertainty in data

- Why uncertainty matters
 - All measurements and models have uncertainty.
 - Ignoring uncertainty can lead to misleading conclusions.
- Sources of uncertainty
 - Measurement error, sampling variability, model assumptions.
- Key visualisation techniques
 - Confidence intervals, error bars, density plots, bootstrapping.

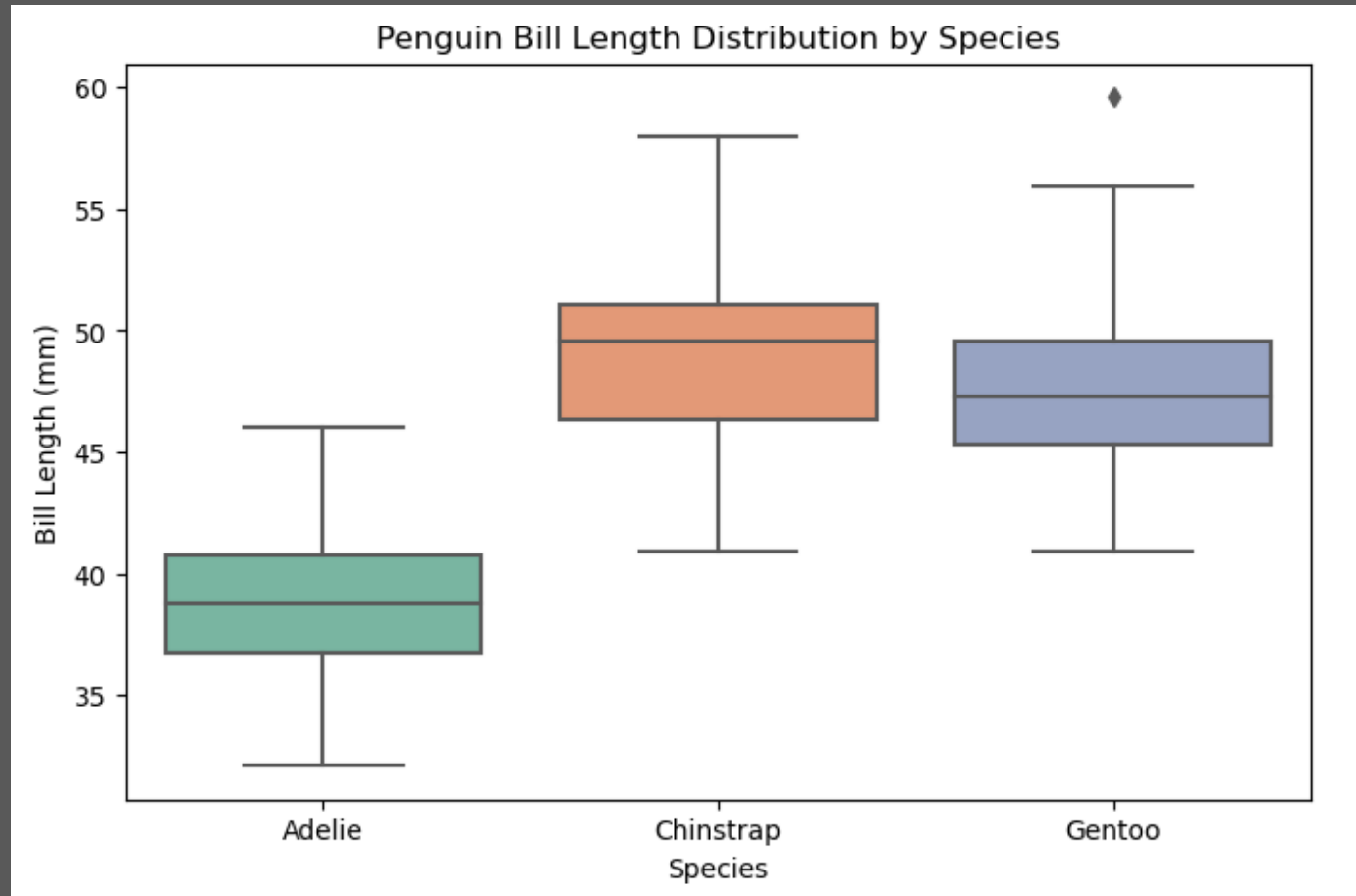
Confidence intervals



Error Bars

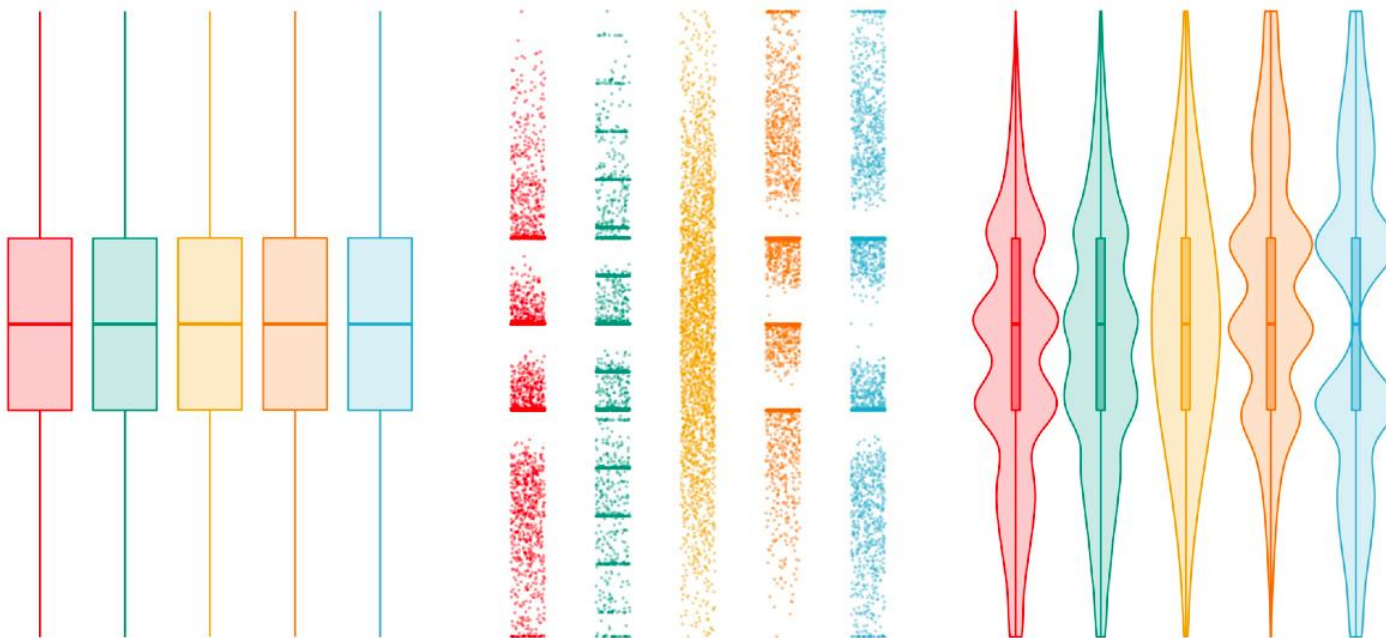


Box plots



Same boxplot, different distributions

Boxplots are great. They show medians and ranges and enable comparison of different groups. However, boxplots can be misleading. Different datasets can have the same descriptive statistics (left), but quite different underlying distributions (middle). Therefore, it is crucial to visualize the distribution in addition to descriptive statistics. Violin plots with integrated boxplots are great for this.



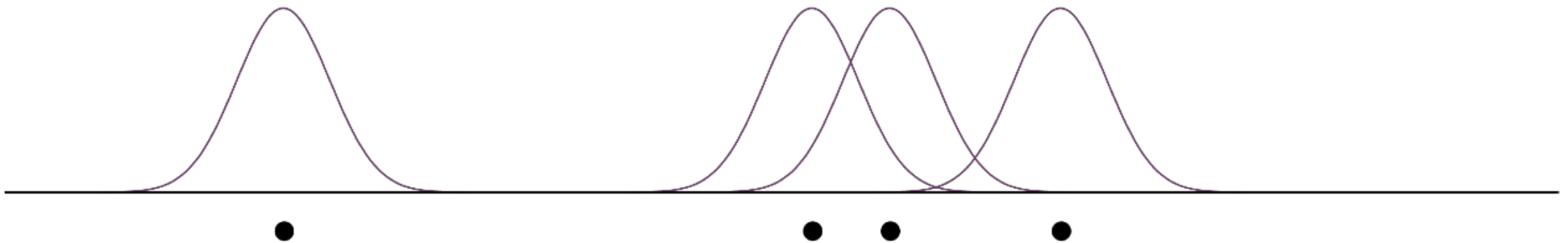
Kernel Density Estimation

- For a set of input data points...



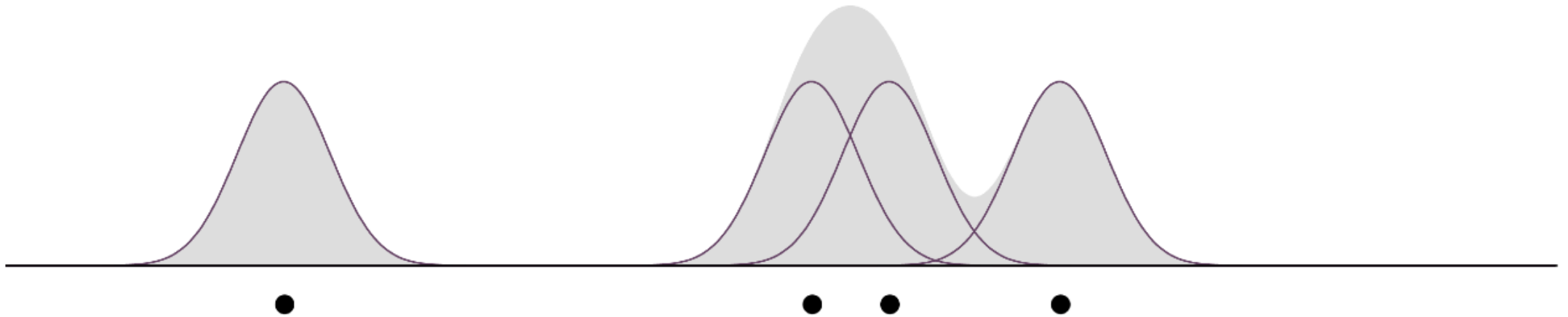
Kernel Density Estimation

- Represent each point with a “kernel” distribution...

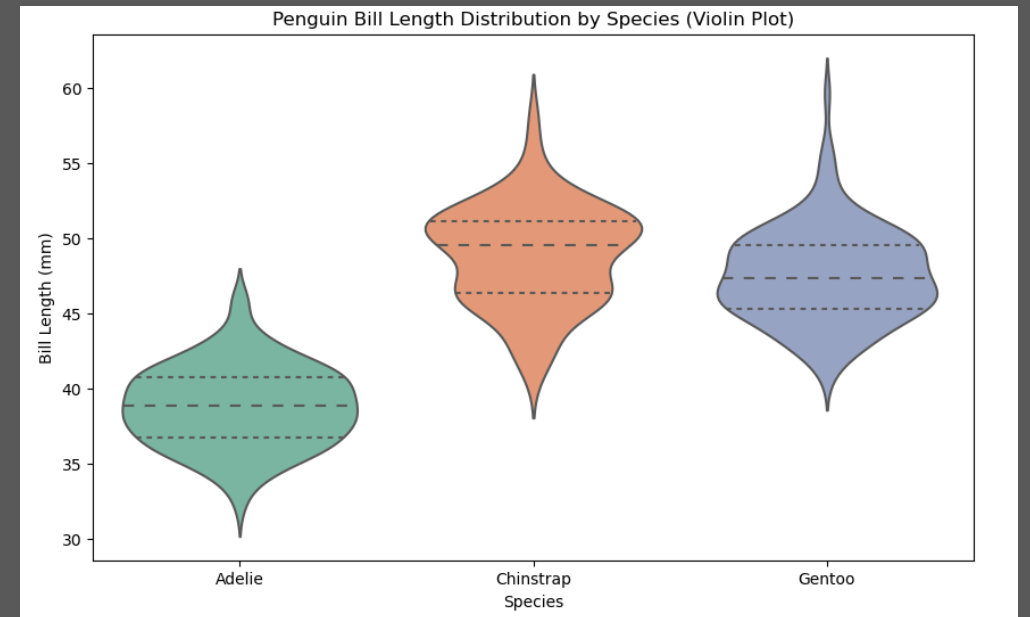
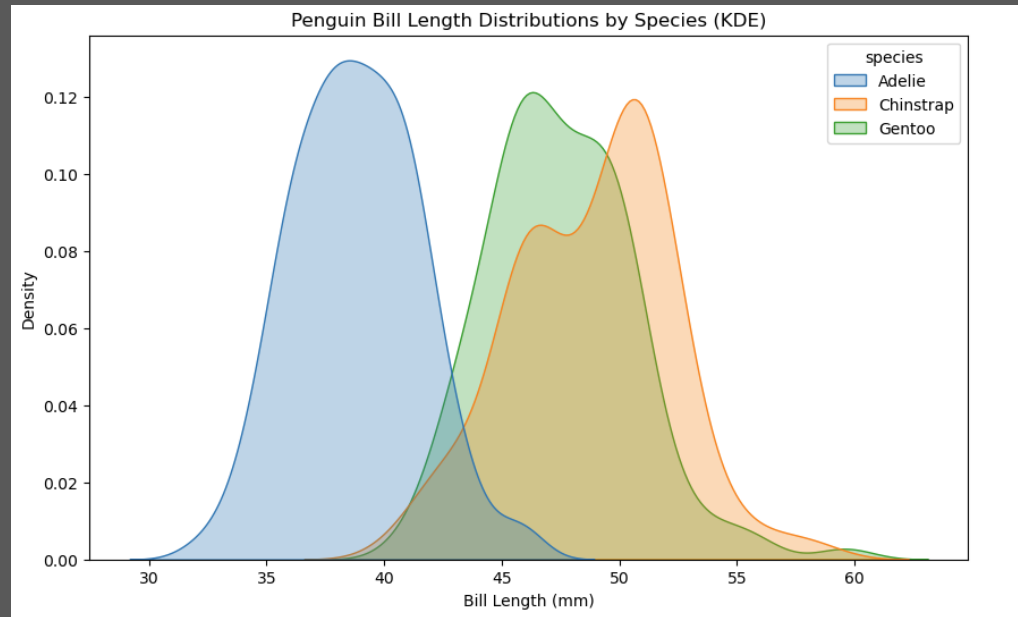


Kernel Density Estimation

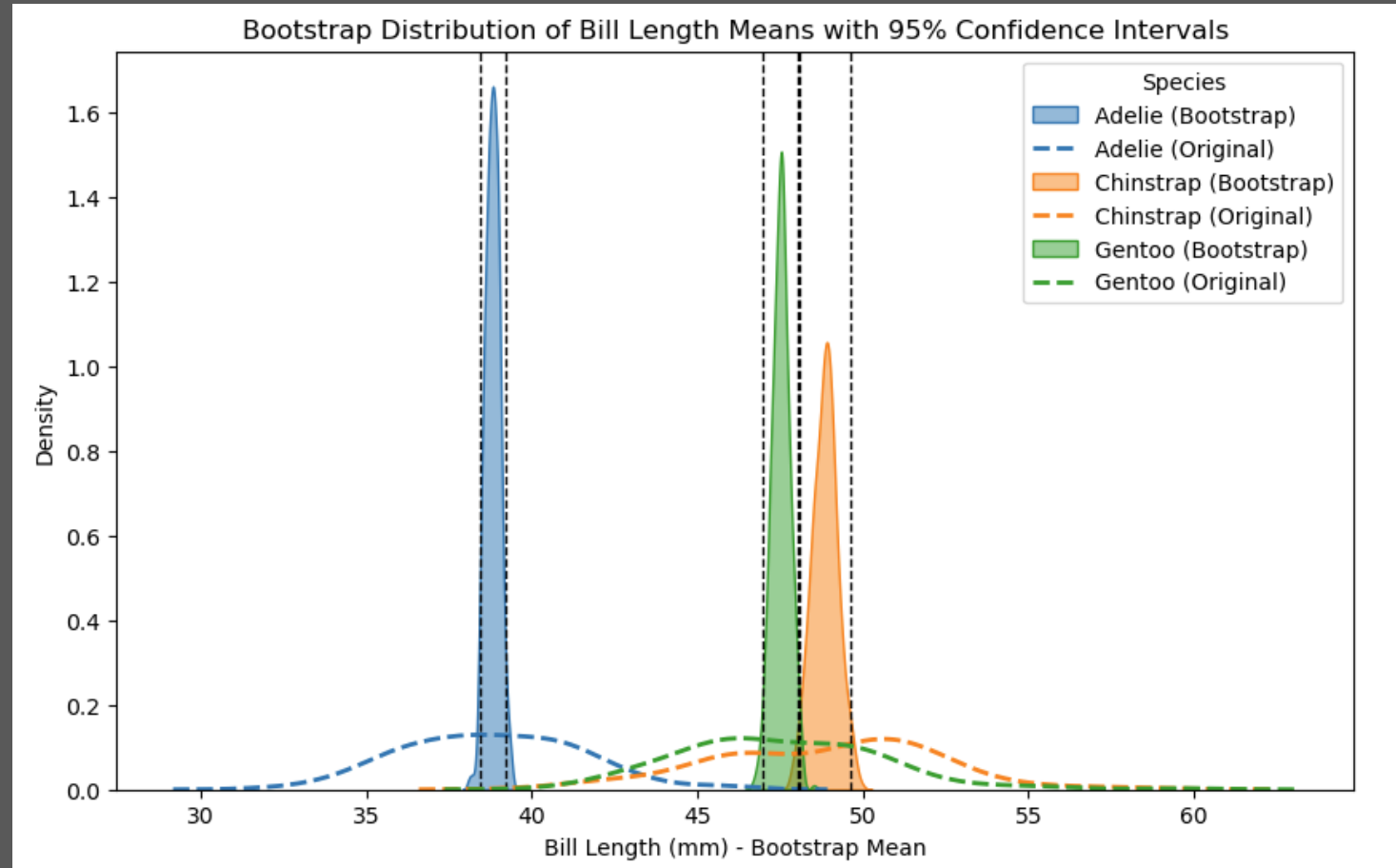
- Sum the kernels to form a density estimate
- Sized by bandwidth (standard deviation)



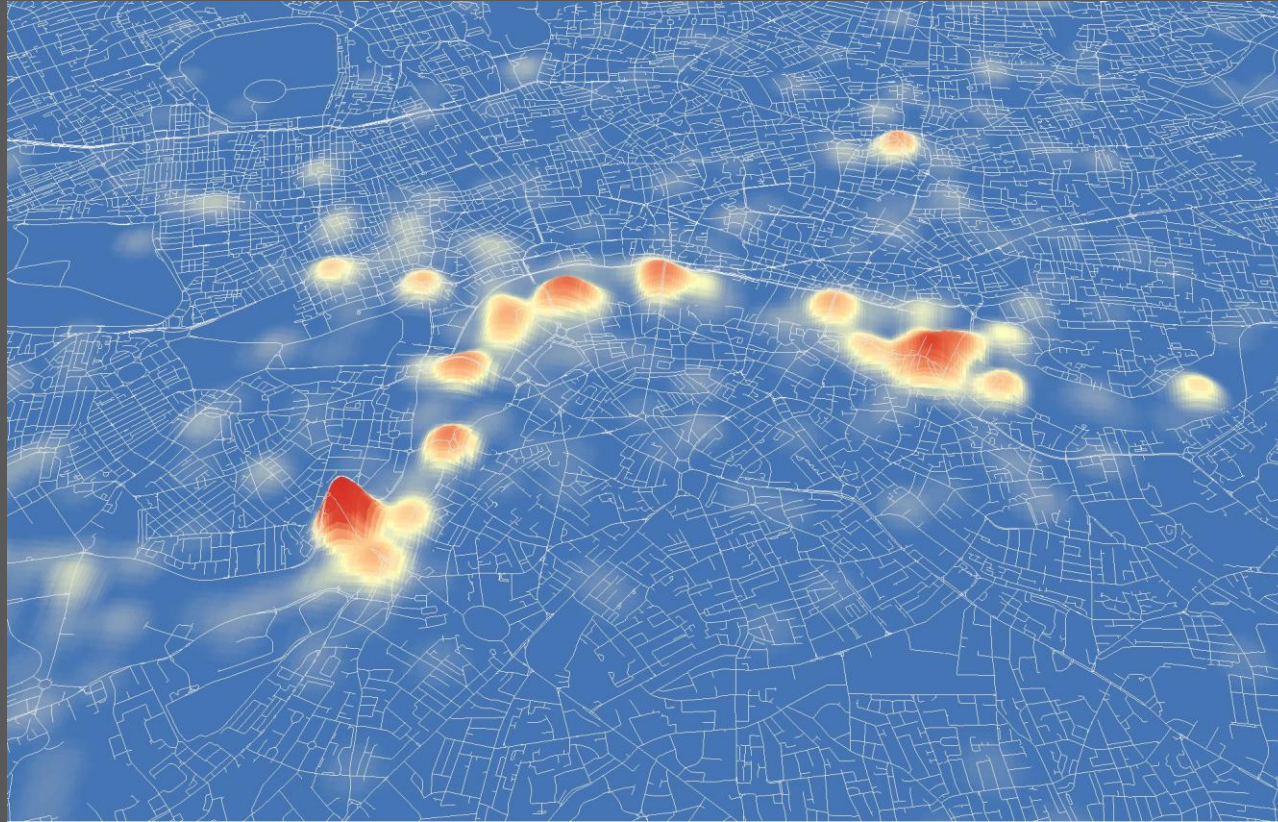
KDE and violin plots



Bootstrapping



Density of social media activity



The Diamond Jubilee in London: A Tweet Location Analysis

Questions?